

Mathematics for Machine Learning

MAD-B3-2526-S2-MAT0611

Bayesian Statistics and Inference

Statistical Paradigms

Frequentist Statistics

- Parameters are fixed but unknown
- Probability = long-run frequency
- Inference based on hypothetical repeated sampling
- Confidence intervals, p-values, hypothesis tests

Bayesian Statistics

- Parameters are random variables with distributions
- Probability = degree of belief
- Inference updates beliefs with data
- Prior $\rightarrow$ Posterior, credible intervals

Bayes' Theorem (Review)

For events A and B :

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

In terms of parameters and data:

$$P(\theta|D) = \frac{P(D|\theta) \cdot P(\theta)}{P(D)}$$

Where:

- θ = parameter(s) of interest
- D = observed data

The Bayesian Framework

$$\text{Posterior} = \frac{\text{Likelihood} \times \text{Prior}}{\text{Evidence}}$$

$$P(\theta|D) = \frac{P(D|\theta) \cdot P(\theta)}{P(D)}$$

Components:

- **Prior** $P(\theta)$: Our belief about θ before seeing data
- **Likelihood** $P(D|\theta)$: Probability of data given parameter
- **Posterior** $P(\theta|D)$: Updated belief after seeing data
- **Evidence** $P(D) = \int P(D|\theta)P(\theta)d\theta$: Normalizing constant

Why the Evidence Often Doesn't Matter

$$P(\theta|D) = \frac{P(D|\theta) \cdot P(\theta)}{P(D)}$$

Since $P(D)$ doesn't depend on θ , we can write:

$$P(\theta|D) \propto P(D|\theta) \cdot P(\theta)$$

"Posterior is proportional to Likelihood times Prior"

We can often work without calculating $P(D)$ and normalize at the end

Bayesian Inference Process

1. Choose a prior distribution $P(\theta)$

- Represents initial beliefs about the parameter

2. Collect data D

- Observations from the process

3. Specify the likelihood $P(D|\theta)$

- Probability model for the data

4. Calculate the posterior $P(\theta|D)$

- Updated beliefs combining prior and data

5. Make inference from the posterior distribution

Example: Coin Flipping

Problem: Is a coin fair? We flip it 10 times and get 7 heads.

Prior: Let θ = probability of heads

- We start with $\theta \sim \text{Beta}(2, 2)$ (slightly favors fair coins)

Likelihood: Given θ , the data follows:

- $D|\theta \sim \text{Binomial}(10, \theta)$
- $P(D|\theta) = \binom{10}{7} \theta^7 (1 - \theta)^3$

Posterior: (we'll derive this next)

Beta Distribution (Quick Review)

$$\text{Beta}(\alpha, \beta) : \quad f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)} \quad \text{for } x \in [0, 1]$$

where $B(\alpha, \beta)$ is the Beta function (normalizing constant).

Properties:

- Mean: $\frac{\alpha}{\alpha+\beta}$
- Mode: $\frac{\alpha-1}{\alpha+\beta-2}$ (for $\alpha, \beta > 1$)
- Flexible shapes depending on α, β

Useful for modeling probabilities

Coin Example: Posterior Calculation

Prior: $\theta \sim \text{Beta}(2, 2)$ so $P(\theta) \propto \theta^{2-1}(1 - \theta)^{2-1} = \theta(1 - \theta)$

Likelihood: $P(D|\theta) \propto \theta^7(1 - \theta)^3$

Posterior:

$$P(\theta|D) \propto P(D|\theta) \cdot P(\theta) \propto \theta^7(1 - \theta)^3 \cdot \theta(1 - \theta)$$

$$P(\theta|D) \propto \theta^8(1 - \theta)^4 = \theta^{9-1}(1 - \theta)^{5-1}$$

Which is proportional to the density of $\text{Beta}(9, 5)$

Posterior: $\theta|D \sim \text{Beta}(9, 5)$

Coin Example: Visualization

Prior: $\text{Beta}(2, 2) \rightarrow$ broad, centered at 0.5

Data: 7 heads out of 10 flips

Posterior: $\text{Beta}(9, 5) \rightarrow$ narrower, centered around 0.64

The posterior:

- Pulls toward the data ($7/10 = 0.70$)
- But is influenced by the prior (which favored 0.5)
- Has less uncertainty than the prior (more peaked)

The data updated our beliefs

Conjugate Priors

A prior is **conjugate** to a likelihood if the posterior has the same distributional form as the prior.

Benefits:

- Analytical solutions (no numerical integration)
- Easy interpretation
- Efficient computation

Common Conjugate Pairs

Likelihood	Parameter	Prior (and Posterior)
Binomial	p	Beta
Normal	μ	Normal
Normal	σ^2	Inverse-Gamma
Poisson	λ	Gamma
Exponential	λ	Gamma
Categorical	$\mathbf{p}$	Dirichlet

[More examples of conjugate priors \(Wikipedia\)](#)

Conjugate Prior: Beta-Binomial

Setup:

- Data: $X \sim \text{Binomial}(n, \theta)$ (observe x successes)
- Prior: $\theta \sim \text{Beta}(\alpha, \beta)$

Posterior:

$$\theta|X \sim \text{Beta}(\alpha + x, \beta + n - x)$$

Interpretation:

- α = prior "pseudo-successes"
- β = prior "pseudo-failures"
- Data adds x successes and $n - x$ failures

Conjugate Prior: Normal-Normal (Known Variance)

Setup:

- Data: $X_1, \dots, X_n \sim N(\mu, \sigma^2)$ with σ^2 known
- Prior: $\mu \sim N(\mu_0, \sigma_0^2)$

Posterior:

$$\mu|X \sim N(\mu_n, \sigma_n^2)$$

Where:

$$\mu_n = \frac{1}{\frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}} \left(\frac{\mu_0}{\sigma_0^2} + \frac{\sum_{i=1}^n x_i}{\sigma^2} \right), \sigma_n^2 = \left(\frac{1}{\sigma_0^2} + \frac{n}{\sigma^2} \right)^{-1}$$

Weighted average of prior mean and sample mean

Prior Selection

Types of Priors:

1. **Informative Prior:** Strong beliefs based on previous knowledge
 - Example: $\theta \sim \text{Beta}(20, 20)$ for a coin
2. **Weakly Informative Prior:** Mild constraints, allows data to dominate
 - Example: $\theta \sim \text{Beta}(2, 2)$
3. **Non-informative (Flat) Prior:** Minimal prior knowledge
 - Example: $\theta \sim \text{Beta}(1, 1) = \text{Uniform}(0, 1)$

Impact of Prior Choice

Weak Data + Strong Prior: Posterior $\approx$ Prior

Weak Data + Weak Prior: Posterior still has high uncertainty

Strong Data + Strong Prior: Battle between prior and data

Strong Data + Weak Prior: Posterior $\approx$ Likelihood

As $n \rightarrow \infty$, the prior influence vanishes (with proper priors)

Bayesian Point Estimates

From the posterior $P(\theta|D)$, we can extract point estimates:

1. Posterior Mean: $E[\theta|D]$

- Minimizes squared error loss

2. Posterior Median: 50th percentile

- Minimizes absolute error loss

3. MAP (Maximum A Posteriori): Mode of posterior

- Most probable value
- $\arg \max_{\theta} P(\theta|D)$

Credible Intervals

A **95% credible interval** is an interval $[a, b]$ such that:

$$P(a \leq \theta \leq b | D) = 0.95$$

Interpretation: Given the data, there's a 95% probability that θ lies in $[a, b]$.

Types:

- **Equal-tailed:** 2.5% in each tail
- **Highest Posterior Density (HPD):** Shortest interval with 95% probability

Credible vs Confidence Intervals

Frequentist Confidence Interval (95%):

- "In repeated sampling, 95% of such intervals contain the true parameter"
- The interval is random; the parameter is fixed
- Cannot say "95% probability θ is in this interval"

Bayesian Credible Interval (95%):

- "Given this data, 95% probability θ is in this interval"
- The parameter is random; the interval is fixed (given data)
- Direct probability statement about the parameter

Example: Normal Mean with Known Variance

Problem: Estimate mean IQ μ from a sample.

- Known: $\sigma = 15$ (population SD)
- Data: $n = 25$ students, $\bar{x} = 108$
- Prior: $\mu \sim N(100, 10^2)$ (mild belief around 100)

Posterior:

$$\sigma_n^2 = \frac{1}{\frac{1}{100} + \frac{25}{225}} = \frac{1}{0.01 + 0.111} = 8.26$$

$$\mu_n = \frac{\frac{100}{100} + \frac{25 \cdot 108}{225}}{\frac{1}{100} + \frac{25}{225}} = \frac{1 + 12}{0.121} = 107.4$$

Posterior: $\mu|D \sim N(107.4, 8.26)$

Example: IQ Inference

Posterior: $\mu|D \sim N(107.4, 8.26)$, so $SD = 2.87$

Point Estimate: $E[\mu|D] = 107.4$

95% Credible Interval:

$$107.4 \pm 1.96 \times 2.87 = [101.8, 113.0]$$

Interpretation: Given our data and prior beliefs, there's a 95% probability that the true mean IQ is between 101.8 and 113.0.

Bayesian Hypothesis Testing

Question: Is a parameter equal to a specific value?

Bayesian Approach: Calculate posterior probability directly

Example: Is the coin fair? ($\theta = 0.5$?)

With posterior $\theta|D \sim \text{Beta}(9, 5)$:

- Calculate $P(\theta > 0.5|D)$
- Calculate $P(0.45 < \theta < 0.55|D)$ (region around fair)

No p-values, no significance levels, just probability

Bayes Factor

Compare two hypotheses: H_1 vs H_0

Bayes Factor:

$$BF = \frac{P(D|H_1)}{P(D|H_0)} = \frac{P(H_1|D)/P(H_1)}{P(H_0|D)/P(H_0)}$$

Interpretation:

- $BF > 1$: Data favors H_1
- $BF < 1$: Data favors H_0
- $BF = 3$ to 10 : Moderate evidence
- $BF > 10$: Strong evidence
- $BF > 100$: Decisive evidence

Bayesian Prediction

Posterior Predictive Distribution: Predict new data $\tilde{X}$

$$P(\tilde{X}|D) = \int P(\tilde{X}|\theta)P(\theta|D)d\theta$$

Interpretation:

- Average predictions over all possible θ values
- Weighted by posterior probability $P(\theta|D)$
- Accounts for parameter uncertainty more directly than point estimates

Example: Posterior Prediction

Coin with $\theta|D \sim \text{Beta}(9, 5)$

Question: What's the probability of heads on the next flip?

$$P(\tilde{X} = 1|D) = \int_0^1 \theta \cdot P(\theta|D) d\theta = E[\theta|D]$$

For $\text{Beta}(9, 5)$: $E[\theta|D] = \frac{9}{9+5} = \frac{9}{14} \approx 0.643$

Prediction: About 64.3% chance of heads on next flip.

Comparing Paradigms

Problem: Estimate coin bias from 7 heads in 10 flips

Frequentist:

- Point estimate: $\hat{\theta} = 0.7$
- 95% CI: $[0.40, 0.90]$ (approximate)
- "If repeated many times, 95% of such intervals contain true θ "

Bayesian (with Beta(2,2) prior):

- Posterior: Beta(9, 5)
- Point estimate: $E[\theta|D] = 0.64$
- 95% credible: $[0.38, 0.86]$
- "95% probability $\theta \in [0.38, 0.86]$ given the data"

When to Use Bayesian Methods?

Bayesian methods excel when:

- You have meaningful prior information
- Sample sizes are small
- You need probability statements about parameters
- Complex hierarchical or multilevel structures
- Sequential data collection (update as data arrives)
- Need to quantify uncertainty in predictions

Both approaches are valuable and valid; choose based on the problem

Example: Medical Diagnosis

Scenario: Test for a rare disease (1% prevalence)

- Test sensitivity: 95% (true positive rate)
- Test specificity: 90% (true negative rate)
- You test positive. What's the probability you have the disease?

Frequentist: Sensitivity and specificity only

Bayesian: Use Bayes' theorem with prevalence as prior

Medical Diagnosis Solution

Let D = has disease, T = tests positive

- $P(D) = 0.01$ (prior/prevalence)
- $P(T|D) = 0.95$ (true positive rate/sensitivity/recall)
- $P(T|no D) = 0.10$ (false positive rate/1 - specificity)

Bayes' Theorem:

$$\begin{aligned} P(D|T) &= \frac{P(T|D) \cdot P(D)}{P(T|D) \cdot P(D) + P(T|no D) \cdot P(no D)} \\ &= \frac{0.95 \times 0.01}{0.95 \times 0.01 + 0.10 \times 0.99} = \frac{0.0095}{0.1085} \approx 0.0876 \end{aligned}$$

Only 8.76% chance you have the disease despite testing positive

Hierarchical Bayesian Models

Structure: Parameters have hyperparameters with their own priors

Example: Students in multiple schools

- Student scores: $y_{ij} \sim N(\mu_j, \sigma^2)$
- School means: $\mu_j \sim N(\mu, \tau^2)$
- Hyperprior: $\mu \sim N(m, s^2)$

Benefits:

- Share information across groups
- Partial pooling (balance between no pooling and complete pooling)
- Better estimates for small groups

Bayesian Computation

When posteriors are complex (no conjugacy):

Markov Chain Monte Carlo (MCMC):

- Gibbs Sampling
- Metropolis-Hastings
- Hamiltonian Monte Carlo (HMC)

Software Tools:

- **Stan**: Modern probabilistic programming language
- **PyMC**: Python library for Bayesian modeling
- **JAGS**: Just Another Gibbs Sampler
- **INLA**: Fast approximate inference